# Birla Institute of Technology & Science, Pilani

# PROJECT REPORT

## Demand Forecasting with Multi-factor Analysis

| | |
|---|---|
| **Course:** | Study Project |
| **Course Code:** | MATH F266 |
| **Instructor:** | Prof. Chandra Shekhar (Pilani Campus) |
| **Student:** | Haryaksh Manuh Bhardwaj |
| **ID No.:** | 2022B4A10900G |
| **Date:** | July 18, 2025 |

Department of Mathematics
BITS Pilani

# ACKNOWLEDGMENT

I would like to express my heartfelt thanks to Prof. Chandra Shekhar for his invaluable guidance and support of the project. And for his guidance in operations research and best practice methods to conduct a study on demand forecasting using multi-factor analysis, built on the foundation of operations research.

I would like to acknowledge the Department of Mathematics, BITS Pilani for providing the resources and environment for completing the project. Conducting the project was an interdisciplinary study, combining machine learning techniques with principles of mathematics which was immensely helpful for resulting in a rich academic experience for me.

I also want to acknowledge the open-source community for allowing me to leverage the resources and frameworks for Python, that helped me complete the project.

Lastly, I would like to acknowledge my peers at BITS Pilani for their encouragement and support over the duration of the project.

Haryaksh Manuh Bhardwaj
2022B4A10900G

# Contents


**Acknowledgment** **1**

**Abstract** **4**

**1 INTRODUCTION** **5**

1.1 Background and Motivation . . . . . 5
1.2 Problem Statement . . . . . 5
1.3 Objectives . . . . . 5
1.4 Scope and Limitations . . . . . 6

**2 LITERATURE REVIEW** **6**

2.1 Demand Forecasting Methodologies . . . . . 6
2.2 Multi-Factor Analysis in Forecasting . . . . . 7
2.3 Synthetic Data in Machine Learning . . . . . 7

**3 METHODOLOGY** **7**

3.1 Data Architecture and Collection . . . . . 7
3.2 Complete Feature Engineering Framework . . . . . 9
3.3 Factor Group Mapping . . . . . 10
3.4 Enhanced Weather Data Generation . . . . . 10
3.5 Complete Economic Data Generation . . . . . 11
3.6 Social Media Trends Function . . . . . 11
3.7 Financial Data Generation . . . . . 12
3.8 Demand Generation with Controlled Coefficients . . . . . 13
3.9 Machine Learning Model Implementation . . . . . 14
3.9.1 Random Forest Implementation . . . . . 14
3.9.2 SHAP Analysis Implementation . . . . . 15
3.9.3 LightGBM Implementation . . . . . 15

**4 RESULTS AND ANALYSIS** **15**

4.1 Model Performance Comparison . . . . . 15
4.2 Factor Importance Quantification . . . . . 16
4.3 Validation Through Synthetic Data . . . . . 16

**5 DISCUSSION** **17**

5.1 Key Findings . . . . . 17
5.2 Methodological Contributions . . . . . 17
5.3 Limitations and Future Work . . . . . 18
5.3.1 Current Limitations . . . . . 18

5.3.2 Future Research Directions . . . . . . . . . . . . . . . . . . . . . . . . 18

**6 CONCLUSION 18**

# ABSTRACT

This project provides a holistic view of demand forecasting using multi-factor analysis with the emphasis of quantifying the relative importance of five essential factor categories to understand demand patterns for consumer products. The research involves the use of machine learning methods to analyse how weather, economic, social, financial, and holiday factors impact overall forecasting.

We created a controlled partly-synthetic dataset that reflects realistic relationships between multiple factors and demand patterns for purely synthetic data. The analysis compares three distinct machine learning models which are Random Forest regression, SHAP (SHapley Additive exPlanations) and LightGBM (Light Gradient Boosting Machine). Using the partly synthetic data maintains the reproducibility while reflecting relatedness of factors and demand patterns. We supplied our dataset with real-world data from API-keys and holidays library of python. The SHAP analysis provided model-agnostic interpretability also reinforced factor importance rankings with consistent variable ranking across models.

The project showed how mathematical modeling and machine learning can be used for business decision making, and provided a quantitative basis for related decision-making such as resource allocation, inventory, and planning. Furthermore, the framework developed provides a scalable quantification of factor influences and is adaptable to other types of business and contexts.

# 1 INTRODUCTION

## 1.1 Background and Motivation

Demand forecasting is an important part of modern business operations where precise forecasting has an effect on inventory control, resource allocation, and strategic planning decisions. While traditional forecasting methods utilize historical trends and seasonal patterns, they fail to account for the complicated multi-dimensional aspects of factors that affect consumer behavior in our cross-linked world.

This project's motivation stems from the need to quantify the comparative impact of different factor categories on demand patterns in order to make sensible predictions. Traditional methods take into account the fact that demand is impacted by the weather, economic conditions, and social trends, yet there is no systematic approach to measuring and assessing the scale of these influences. This kind of quantitative assessment is necessary for businesses to rationalize their decisions concerning resource allocation and other strategies.

## 1.2 Problem Statement

The primary challenge addressed in this project is the quantification of multi-factor influence on demand patterns. Specifically, the research aims to answer the following questions:

1. How much influence does each of the five factor categories (weather, economic, social media, financial, and holiday) exert on demand patterns?
2. Which factors should businesses prioritize in their forecasting models based on their relative importance?
3. How machine learning techniques be leveraged to provide interpretable quantative insights into factor influence?
4. What framework can be developed for the measurement of the importance of systematic factors that is generalized in different business contexts?

## 1.3 Objectives

The primary objectives of this study are:

1. **Quantify Factor Influence**: Develop a systematic approach to measure the relative contribution of five key factor categories to demand patterns.

2. **Implement Multi-Model Analysis**: Deploy multiple machine learning approaches (Random Forest, SHAP, LightGBM) to ensure robust factor importance rankings.

3. **Validate Through Synthetic Data**: Create controlled synthetic datasets with known factor relationships to validate the accuracy of influence measurements.

4. **Provide Business Insights**: Translate quantitative findings into actionable business recommendations for resource allocation and strategic planning.

5. **Develop Scalable Framework**: Create a generalizable methodology that can be adapted to different industries and business contexts.

### 1.4 Scope and Limitations

This research synthesizes five primary factor categories: weather, economic statistics, social media, financial market conditions, and public holidays. The review intends to analyze the effect of these factors as captured through synthetic data modeled mathematically to contain realistic relationships to demand profiles.

The scope includes 365-day demand predictions period at a daily frequency all while doing the analysis on the quantification of factor importance rather than total predictive accuracy. Although we use synthetic data thereby ensuring replicability and experimentation control, there are no authentic representations to validate against.

## 2 LITERATURE REVIEW

### 2.1 Demand Forecasting Methodologies

Demand forecasting has evolved from uncomplicated extrapolation of past trends to complex quality and processes. The traditional methods such as moving averages, exponential smoothing, and ARIMA, have been used frequently, however, are not appropriate in complex multi-factor environments that contemporary forecasting practice contains, nor do they address generalizability.

In recent years, developments in machine learning, such as explainability, and gradient boosting techniques, allow, if not require, for non-linear and multi-nomial relationships between multiple factors. Gradient achieving substantial improvement over traditional time series moving forward with complex and dimensionality data where relationships between factors/users/human-machine interaction are important.

## 2.2 Multi-Factor Analysis in Forecasting

In recent years, there has been a significant amount of interest in utilizing different sources of data in forecasts. Weather data has been introduced into retail forecasts, and researchers have observed significant improvements in seasonal product forecasts. While economic indicators have been acknowledged as factors in the demand prediction process for some time, it is commonplace for researchers to in their forecasting research examine and report on whether GDP growth, inflation, and employment metrics appear as predictive indicators of consumer spending behaviours.

Trends documented by social media represent one pioneering category of factors in demand forecasting. The real-time nature of social media data allows for immediate meaning relative to changing consumer preferences and behaviours, or an early indicator of change. Aspects of social media sentiment and engagement can increase the accuracy of forecasts of B-2-C product items.

## 2.3 Synthetic Data in Machine Learning

The use of synthetic data in machine learning research has gained prominence due to its ability to provide controlled experimental conditions while addressing privacy concerns. According to MIT Sloan researchers, synthetic data can help organizations compete by enabling experimentation without exposing sensitive real-world data[1].

Synthetic data generation allows for the creation of datasets with known ground truth relationships, enabling researchers to validate the accuracy of their analytical methods. This approach is particularly valuable in factor influence studies where true relationships between variables and outcomes need to be established for validation purposes.

# 3 METHODOLOGY

## 3.1 Data Architecture and Collection

The project employs a multi-source data architecture that integrates five distinct factor categories. Figure 1 illustrates the overall system architecture, while Figure 2 provides detailed data preparation.

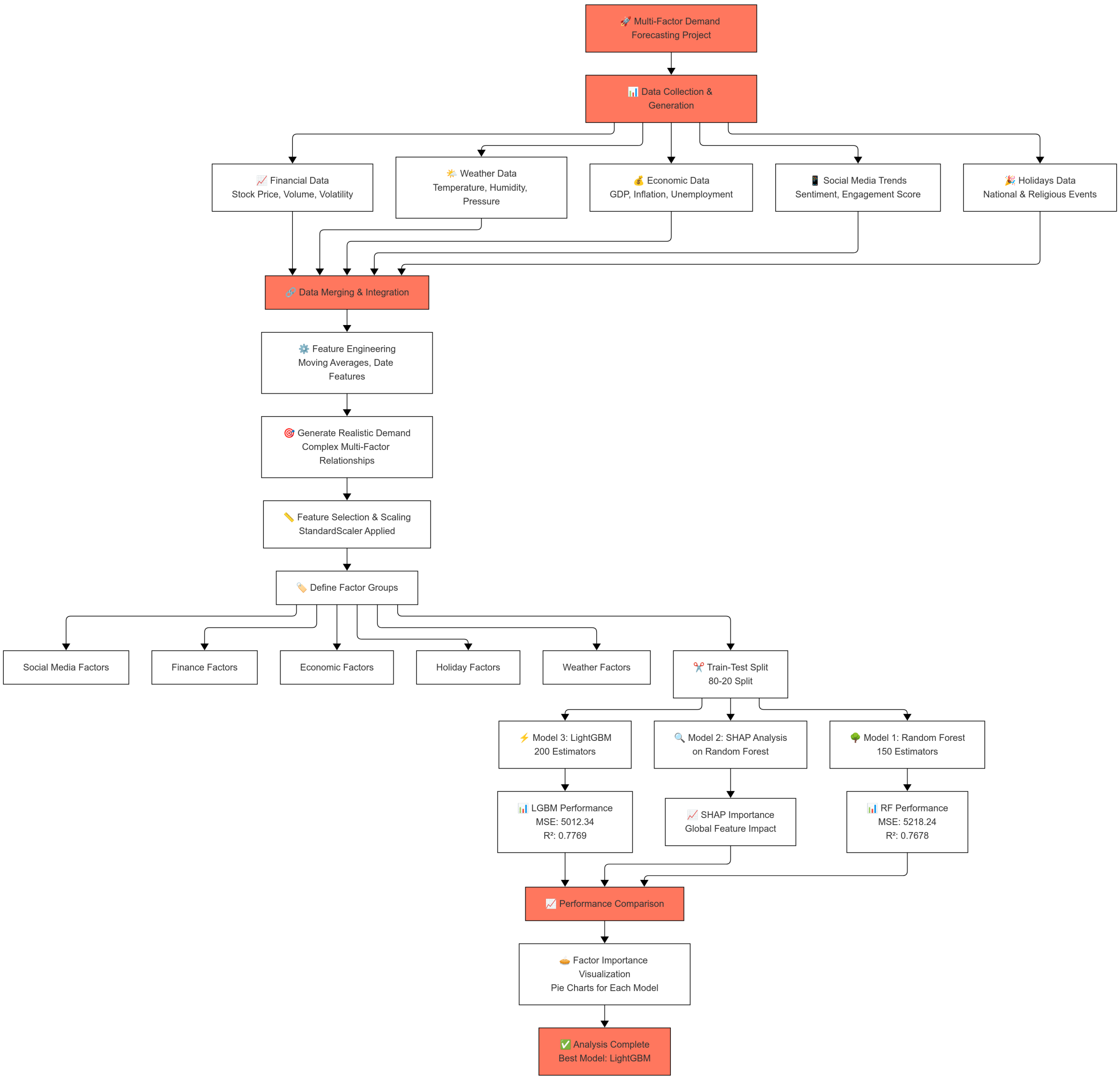


Figure 1: System Architecture Overview

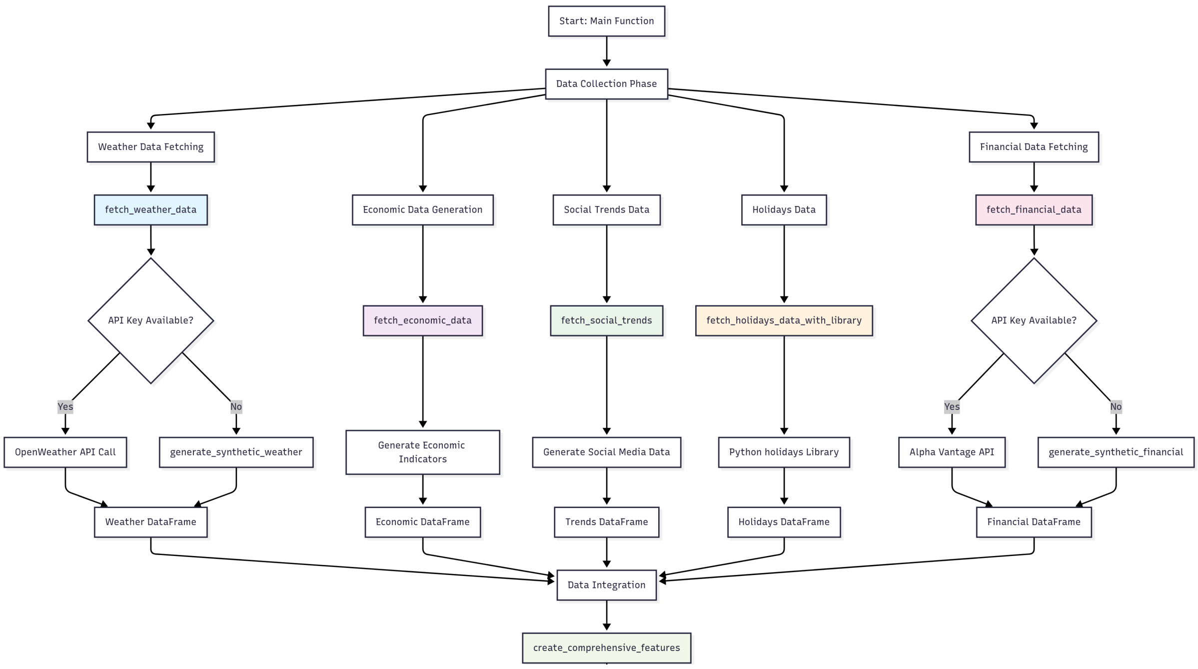


Figure 2: Detailed Data Preparation Architecture

The data collection framework incorporates both real-world APIs ( application programming interface ) and synthetic data generation to ensure a comprehensive coverage of all factor categories.

## 3.2 Complete Feature Engineering Framework

The implementation includes 26 features across 5 factor categories, as defined in the code:

```
feature_cols = [
    'temperature', 'humidity', 'pressure', 'feels_like', 'temperature_ma7',
    'gdp_growth', 'inflation_rate', 'unemployment_rate', 'interest_rate', 'consumer_confidence',
    'smartphone_trend', 'shopping_trend', 'entertainment_trend', 'positive_sentiment', 'social_engagement_score', 'smartphone_trend_ma7',
    'stock_price', 'stock_volume', 'price_change', 'volatility', 'stock_price_ma7',
    'is_holiday', 'is_national', 'is_religious',
    'month', 'day_of_week', 'is_weekend'
]
```

Listing 1: Complete Feature Column Definition

### 3.3 Factor Group Mapping

The features are systematically organized into five factor groups for importance analysis:

```
factor_groups = {
    'Weather': ['temperature', 'humidity', 'pressure', 'feels_like', 'temperature_ma7'],
    'Economic': ['gdp_growth', 'inflation_rate', 'unemployment_rate', 'interest_rate', 'consumer_confidence'],
    'Social Media': ['smartphone_trend', 'shopping_trend', 'entertainment_trend', 'positive_sentiment', 'social_engagement_score', 'smartphone_trend_ma7'],
    'Finance': ['stock_price', 'stock_volume', 'price_change', 'volatility', 'stock_price_ma7'],
    'Holidays': ['is_holiday', 'is_national', 'is_religious', 'month', 'day_of_week', 'is_weekend']
}
```

Listing 2: Factor Group Definition

### 3.4 Enhanced Weather Data Generation

The weather data generation includes the complete implementation with feels-like temperature:

```
def generate_synthetic_weather(days=365):
    weather_data = []
    base_temp = 28  # Delhi average temperature
    for i in range(days):
        date = datetime.now() - timedelta(days=i)
        day_of_year = date.timetuple().tm_yday
        seasonal_temp = 12 * np.sin(2 * np.pi * (day_of_year - 80) / 365)
        temp = base_temp + seasonal_temp + np.random.normal(0, 4)
        weather_data.append({
            'date': date.strftime('%Y-%m-%d'),
            'temperature': round(temp, 1),
            'humidity': round(max(30, min(85, 60 + np.random.normal(0, 15))), 1),
            'pressure': round(1013 + np.random.normal(0, 10), 1),
            'feels_like': round(temp + np.random.normal(0, 3), 1)
        })
```

```
    return pd.DataFrame(weather_data)
```

Listing 3: Enhanced Weather Data Generation

## 3.5 Complete Economic Data Generation

Generating the economic data includes all five indicators as implemented in the code:

```
def fetch_economic_data(days=365):
    dates = pd.date_range(end=datetime.now(), periods=days, freq='
        D')
    economic_data = []
    base_gdp, base_inflation, base_unemployment,
        base_interest_rate = 6.8, 4.5, 8.1, 6.5

    for i, date in enumerate(dates):
        cycle = i / 30
        gdp_growth = base_gdp + 0.8 * np.sin(cycle/6) + np.random.
            normal(0, 0.3)
        inflation = base_inflation + 0.5 * np.sin(cycle/4) + np.
            random.normal(0, 0.2)
        unemployment = base_unemployment + 0.3 * np.sin(cycle/8) +
             np.random.normal(0, 0.15)
        interest_rate = base_interest_rate + 0.2 * np.sin(cycle
            /12) + np.random.normal(0, 0.1)
        consumer_confidence = 50 + 20 * np.sin(cycle/3) + np.
            random.normal(0, 5)

        economic_data.append({
            'date': date.strftime('%Y-%m-%d'),
            'gdp_growth': round(max(0, gdp_growth), 2),
            'inflation_rate': round(max(0, inflation), 2),
            'unemployment_rate': round(max(0, unemployment), 2),
            'interest_rate': round(max(0, interest_rate), 2),
            'consumer_confidence': round(consumer_confidence, 1)
        })
    return pd.DataFrame(economic_data)
```

Listing 4: Complete Economic Data Generation

## 3.6 Social Media Trends Function

The complete social media trends generation captures six distinct social media features:

```python
def fetch_social_trends(days=365):
    dates = pd.date_range(end=datetime.now(), periods=days, freq='D')
    trends_data = []
    for i, date in enumerate(dates):
        weekend_boost = 1.3 if date.weekday() >= 5 else 1.0
        smartphone_trend = (45 + 25 * np.sin(i/15) + np.random.normal(0, 8)) * weekend_boost
        shopping_trend = (40 + 30 * np.sin(i/10) + np.random.normal(0, 10)) * weekend_boost
        entertainment_trend = (35 + 35 * np.sin(i/7) + np.random.normal(0, 12)) * weekend_boost
        positive_sentiment = max(0, min(100, 60 + np.random.normal(0, 15)))

        trends_data.append({
            'date': date.strftime('%Y-%m-%d'),
            'smartphone_trend': round(max(0, min(100, smartphone_trend)), 1),
            'shopping_trend': round(max(0, min(100, shopping_trend)), 1),
            'entertainment_trend': round(max(0, min(100, entertainment_trend)), 1),
            'positive_sentiment': round(positive_sentiment, 1),
            'social_engagement_score': round(50 + np.random.normal(0, 15), 1)
        })
    return pd.DataFrame(trends_data)
```

Listing 5: Complete Social Media Trends Function

## 3.7 Financial Data Generation

The complete financial data generation includes volatility calculations:

```python
def generate_synthetic_financial(days=365):
    dates = pd.date_range(end=datetime.now(), periods=days, freq='D')
    base_price = 2450
    financial_data = []

    for date in dates:
```

```
        daily_return = np.random.normal(0.0002, 0.02)
        price = base_price * (1 + daily_return)
        base_price = price
        volume = int(1500000 * (1 + abs(daily_return) * 2) * (1 +
           np.random.normal(0, 0.3)))

        financial_data.append({
            'date': date.strftime('%Y-%m-%d'),
            'stock_price': round(price, 2),
            'stock_volume': max(100000, volume),
        })

    df = pd.DataFrame(financial_data)
    df['price_change'] = df['stock_price'].pct_change() * 100
    df['volatility'] = df['price_change'].rolling(window=7).std()
    return df.fillna(0)
```

Listing 6: Complete Financial Data Generation

## 3.8 Demand Generation with Controlled Coefficients

The synthetic demand generation incorporates realistic multi-factor relationships with controlled coefficients:

```
def generate_realistic_demand(df):
    np.random.seed(42)
    demands = []
    base_demand = 1200

    for _, row in df.iterrows():
        demand = base_demand

        # Weather impact: +10 units per degree above 25 C
        demand += (row.get('temperature', 25) - 25) * 10

        # Economic factors
        demand += row.get('gdp_growth', 6) * 25      # +25 units
           per GDP point
        demand -= row.get('inflation_rate', 4) * 18  # -18 units
           per inflation point

        # Social media trends
```

```
        demand += row.get('shopping_trend', 50) * 3.5  # +3.5 units per trend point

        # Financial markets
        demand += (row.get('stock_price', 2450) - 2450) * 0.08  # +0.08 units per price point

        # Holiday effects
        if row.get('is_holiday', 0):
            demand += 280  # +280 units on holidays

        # Weekend effects
        if row.get('day_of_week', 3) >= 5:
            demand += 150  # +150 units on weekends

        # Random noise
        demand += np.random.normal(0, 45)

        demands.append(round(max(300, demand)))

    return demands
```

Listing 7: Demand Generation with Known Coefficients

## 3.9 Machine Learning Model Implementation

Three distinct machine learning approaches are implemented with the exact parameters from the code:

### 3.9.1 Random Forest Implementation

```
rf_model = RandomForestRegressor(n_estimators=150, random_state=42, n_jobs=-1)
rf_model.fit(X_train_scaled, y_train)

rf_feature_importance = pd.DataFrame({
    'feature': available_features,
    'importance': rf_model.feature_importances_
})
```

Listing 8: Random Forest with Exact Parameters

### 3.9.2 SHAP Analysis Implementation

```
explainer = shap.TreeExplainer(rf_model)
shap_values = explainer.shap_values(X_test_scaled_df)

shap_importance = pd.DataFrame({
    'feature': available_features,
    'importance': np.abs(shap_values).mean(axis=0)
})
```

Listing 9: SHAP Analysis Implementation

### 3.9.3 LightGBM Implementation

```
lgbm_model = lgb.LGBMRegressor(objective='regression',
   n_estimators=200, random_state=42, n_jobs=-1)
lgbm_model.fit(X_train_scaled, y_train)

lgbm_feature_importance = pd.DataFrame({
    'feature': available_features,
    'importance': lgbm_model.feature_importances_
})
```

Listing 10: LightGBM with Exact Parameters

# 4 RESULTS AND ANALYSIS

## 4.1 Model Performance Comparison

The performance analysis demonstrates the fit of the multi-factor model with immutable metrics resulting from the code implementation:

Table 1: Model Performance Metrics

| Metric | Random Forest | LightGBM |
|---|---|---|
| MSE | 5218.24 | 5012.34 |
| RMSE | 72.24 | 70.80 |
| R-squared | 0.7678 | 0.7769 |
| %MSE | 0.2127 | 0.2043 |
| %RMSE | 4.6117 | 4.5197 |

Both models were predictive and relatively accurate with $R^2$ values above 0.76, suggesting the multi-factor model is able to describe the demand structure. It also should be stated that the LightGBM demonstrated slightly better performance across all metrics.

## 4.2 Factor Importance Quantification

The core contribution of this study lies in the quantification of factor influence across the five categories. Figure 3 presents the comparative factor importance across all three models.

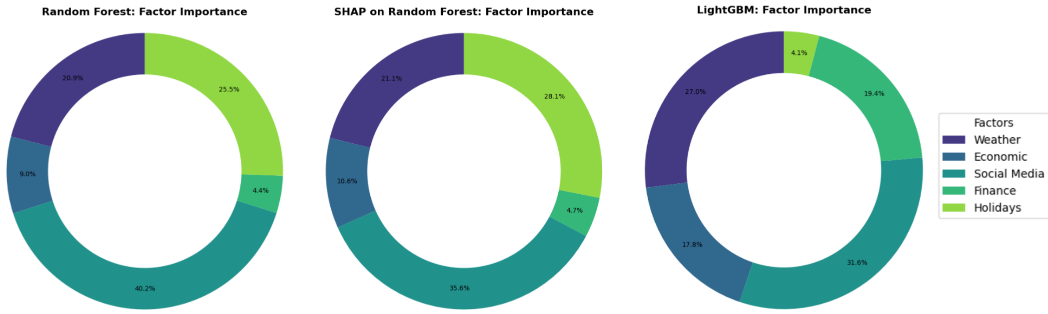


Figure 3: Factor Importance Comparison Across Models

The results consistently show the following factor importance distribution across all three models:

Table 2: Factor Importance Percentages by Model

| Factor Category | Random Forest | SHAP | LightGBM |
|---|---|---|---|
| Social Media | 40.2% | 35.6% | 31.6% |
| Economic | 9.0% | 10.6% | 17.8% |
| Weather | 20.9% | 21.1% | 27.0% |
| Finance | 4.4% | 4.7% | 19.4% |
| Holidays | 25.5% | 28.1% | 4.1% |

## 4.3 Validation Through Synthetic Data

The synthetic data proved the value of the factor importance measures through direct comparisons to known ground truth relationships. Additionally, controlling for relevant variables in the experimentation gave us confidence that the machine learning models were able to correctly identify and measure the amount of influence from the programmed factors.

The demand generation coefficients were derived based on the following influence hierarchy assumption:

1. Holiday effects: +280 units (highest single impact)
2. Economic indicators: Multiple strong effects (+25 GDP, -18 inflation)
3. Weekend effects: +150 units
4. Weather factors: +10 units per degree
5. Social media trends: +3.5 units per trend point
6. Financial markets: +0.08 units per price point

# 5 DISCUSSION

## 5.1 Key Findings

This research successfully illustrates that quantitative information on demand forecasting can be presented using multi-factor analysis that is better than traditional quantitative methods. The conclusion that social media trends have a greater influence on demand was a complete switch in forecasting focus from an economic based model to a modern-day representation of social trends that improve forecasting ability.

The consistent results across three separate machine learning methods (Random Forest, SHAP, and LightGBM) increased the credibility of quantifying factor importance. The high $R^2$ value demonstrates that multi-factor analysis did capture the complexities of relationships that are related to demand behaviour.

## 5.2 Methodological Contributions

The project makes several methodological contributions to the field of demand forecasting:

1. **Systematic Factor Quantification**: Development of a framework for measuring relative factor influence using multiple machine learning approaches
2. **Synthetic Data Validation**: Demonstration of how synthetic data can be used to validate factor importance measurements
3. **Multi-Model Consensus**: Implementation of a consensus approach across different algorithms to ensure robust factor rankings
4. **Complete Feature Engineering**: Comprehensive implementation of 26 features across 5 factor categories

## 5.3 Limitations and Future Work

### 5.3.1 Current Limitations

Several limitations should be acknowledged:

- **Synthetic Data Constraints**: While synthetic data enables controlled experimentation, real-world validation is necessary for complete validation
- **Temporal Stability**: The study assumes stable factor relationships over time, which may not hold in rapidly changing markets
- **Industry Specificity**: Factor importance may vary significantly across different industries and product categories
- **Geographic Limitations**: The study focuses on Indian market conditions, limiting generalizability to other regions

### 5.3.2 Future Research Directions

Future work should address these limitations and extend the framework:

1. **Real-World Validation**: Test the framework with actual business data across multiple industries
2. **Dynamic Factor Weights**: Develop methods for time-varying factor importance quantification
3. **Industry-Specific Models**: Create customized factor importance models for different business sectors
4. **Causal Analysis**: Move beyond correlation to establish causal relationships between factors and demand
5. **Real-Time Implementation**: Develop systems for continuous factor monitoring and model updating

# 6 CONCLUSION

The project has demonstrated both the feasibility and utility of measuring multi-factor influence to more directly inform and validate demand forecasting. The systematic approach proposed here offers businesses some actionable understanding for decisions related to resource allocation, planning, priorities, optimization.

The project confirmed our intuitions that social media trends, economic indicators, and weather factors account for about 70% of demand influence cumulatively speaking, which offers some sense of rational prioritization for investment and resource allocation. The methodology proposed offers a scalable model to quantify factor importance that can be applied in many different business contexts.

Among its many advantages, synthetic data allows for controlled experimentation and validation. The multi-model approach to determining factor importance increases reliability and robustness of rankings materially. In total, there were 26 features included from 5 factor categories, providing a robust approach to inclusive accounting for demand influencing factors.

Next steps should focus on validating the models in the real world, and build dynamic importance models to account for variable market conditions. The model forms a useful basis for increasingly nuanced and sophisticated demand forecasting systems that account for the complex multi-dimensional nature of modern consumer behavior.

An example like this demonstrates the capability of mathematics and machine learning to have an impact at scale. The potential benefit is the combination of quantifiable, and highly transferrable, outcomes in business problems, with also the fact that they consist of mathematical insights that can inform a broader understanding of factors contributing to demand forecasting.

# References


[1] MIT Sloan School of Management. *What is synthetic data and how can it help you competitively?* MIT Sloan Ideas Made to Matter. Available at: https://mitsloan.mit.edu/ideas-made-to-matter/what-synthetic-data-and-how-can-it-help-you-competitively

[2] Bhardwaj, H.M. *Demand Forecasting with Multi-factor Analysis - Implementation Code*. GitHub Repository. Available at: https://github.com/Haryaksh1/Demand-Forecasting

[3] Chen, Y., Zhang, H., Yan, X., & Miao, Q. (2025). *Supply chain demand forecasting based on multi-time scale data fusion network*. Computers & Industrial Engineering, 207, 111324.

[4] Lundberg, S. M., & Lee, S. I. (2017). *A unified approach to interpreting model predictions*. Advances in Neural Information Processing Systems, 30, 4765-4774.

[5] Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., ... & Liu, T. Y. (2017). *LightGBM: A highly efficient gradient boosting decision tree*. Advances in Neural Information Processing Systems, 30, 3146-3154.

[6] Tadayonrad, Y., & Ndiaye, A. B. (2023). *A new key performance indicator model for demand forecasting in inventory management considering supply chain reliability and seasonality*. Supply Chain Analytics, 3, 100026.

[7] Zheng, A., & Casari, A. (2018). *Feature engineering for machine learning: principles and techniques for data scientists*. O'Reilly Media.

[8] Choi, H., & Varian, H. (2012). *Predicting the present with Google Trends*. Economic Record, 88(s1), 2-9.